GridDS will help analyse the data generated by smart meters in the US

In the US, 102.9 million smart meters have been deployed as of 2020. These gadgets record and report electric consumption, voltage, and current to users and grid operators. By 2050, it is anticipated that both the number of smart meters and the amount of energy consumed will have increased by 50 per cent. Although large-scale data gathering and storage have been made possible by energy standards, leveraging this data to reduce costs and consumer demand has been a continuous focus of energy research.

A team from the Lawrence Livermore National Laboratory (LLNL) has created GridDS, an open source data-science toolkit for power and data engineers, to help make the most of all this data. It offers an integrated infrastructure for storing and augmenting energy data as well as a comprehensive set of cutting-edge machine learning models.

The Grid Modernization Lab Consortium (GMLC) of the US Department of Energy provides funding for GridDS. GridDS will help increase the effectiveness of distributed energy resources like smart meters, batteries, and solar photovoltaic panels by offering an integrative software platform to train and evaluate machine learning models. In order to estimate energy consumption and identify impending grid failures, GridDS is also built to make use of sophisticated metering infrastructure, outage management system data, supervisory control data collection, and geographic information systems.

For these many data streams, GridDS has a Python software package that is modular and generalizable. GridDS offers a variety of distinct functions not yet implemented in advanced distribution management systems, which often have very specialised software infrastructure by design. It adapts to dissimilar data sets recorded by various devices.

format that cannot easily be converted to a database, it can be difficult to search through their contents. In fact, according to IDC, by 2025, 80 per cent of all data will be unstructured globally.

Scientists and organisations can already search through mounds of unstructured data thanks to IBM Research's Deep Search. But now the company has launched Deep Search for Scientific Discovery (DS4SD), an open source toolbox for scientific research, to make deep search even more adaptable and accessible. The release of DS4SD represents the next step towards IBM Research's ultimate aim of creating an Open Science Hub for Accelerated Discovery after the release of the Generative Toolkit for Scientific Discovery (GT4SD) in March this year.

Data scientists and engineers have access to the new toolbox through IBM Research's Python package, which interacts and integrates with current services. Developer community contributions are welcomed since the toolkit is open source.

Deep Search is an AI-powered method of gathering, converting, curating, and finally searching through large document collections for information that is too specialised for standard search engines to handle. It then processes these papers using specialised natural language processing, computer vision, and machine learning techniques to produce searchable knowledge graphs. The generated data sets can be used to assist firms in developing models and locating significant trends that guide their choices.

Catalyst and CERT NZ assist the open source community to develop a Samba update

The open source community, Catalyst, and CERT NZ have worked together to develop a significant update for Samba that increases security. Like Microsoft Active Directory, Samba is an open source software package that serves as an active directory domain controller. The Samba project is a member of the Software Freedom Conservancy, and Samba is free software covered by the GNU General Public License. Samba has been offering safe quick file and print services since 1992 for all clients implementing the SMB/CIFS protocol, including all DOS and Windows versions, OS/2, Linux, and many more.

To seamlessly integrate Linux/UNIX servers and desktops into Active Directory systems, the software is a crucial element. It can perform the duties of an ordinary domain member or a domain controller. A Samba upgrade developed in collaboration between CERT NZ and New Zealand's Catalyst will help users everywhere. The potential to help secure Samba was viewed by the organisation as being advantageous to both CERT NZ and others, according to Pieter Meirsman, team head of systems and security at CERT NZ.